

Reachability Freshness Is Consumer-Relative and Witnessed: Sealed Measurement of Quiescence-Monitor Vacuity in Interdomain and Link-State Routing

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Abstract

Routing systems monitor *reachability freshness*: whether the control plane has converged, whether a route is stable, whether a prefix is still validly originated. Each such monitor is a *certificate*. We show, and measure, that quiescence-style monitors—“no update seen for T seconds, therefore converged”—are *consumer-relative vacuous*: the false-clear rate depends on the observer’s vantage and the protocol’s opacity, and a monitor that infers quiescence from update silence clears while the network is still churning. We formalize the false-clear rate against an independent *witness*—a BGP Monitoring Protocol (BMP) feed, a receiver-side adjacency-in change-point, or a dataplane probe—and measure it under a sealed-preregistration protocol. Across control-plane substrates the 60 s quiescence certificate false-clears at 0.351 (BGP), 0.184 (IS-IS), and 0.083 (OSPF), an ordering set by protocol opacity; a posterior detector holds a dominance band at 30–120 s. **[TODO/D8-V1: one-sentence dataplane headline: the D8-V1 join confirms the band and the consumer-relative vacuity on live reachability.]**

Index Terms

BGP, IS-IS, OSPF, route monitoring, convergence, BMP, RPKI, reachability, pre-registration, age of information.

I. INTRODUCTION

The routing instance of the program: a reachability certificate (converged / stable / validly-originated) that ages, is graded against a witness (BMP, the receiver’s adjacency-in, or a dataplane probe), and has a consumer-relative false-clear rate ordered by protocol opacity. Contributions: (i) the framework specialized to routing; (ii) sealed control-plane cells (below); (iii) **[TODO/D8-V1: the D8-V1 dataplane-join cell on live RIPE Atlas reachability];** (iv) **[TODO/D8-V1: the confirmed posterior-detector dominance band as the product surface]**. This paper shares the certificate/witness/refresh-floor grammar with the radio-access and database instances of the program [1].

II. CERTIFICATES, CONSUMERS, WITNESSES

Definition 1 (Quiescence certificate). *An assertion that the routing state is converged/stable, inferred from the absence of updates over a window T at an observer’s vantage.*

Definition 2 (Witness). *An independent signal of actual churn: a BMP route-monitoring feed (RFC 7854) [2], the receiver-side adjacency-in change-points at the consumer router, or a dataplane reachability probe. The witness reports what the quiescence certificate can only infer.*

The false-clear rate is consumer-relative: it depends on vantage (a distant observer sees fewer updates) and on protocol opacity (link-state floods differ from path-vector churn). Route-flap damping [3] and origin validation [4] are the deployed witnessed corrections for two adjacent certificates.

III. SEALED-PREREGISTRATION MEASUREMENT

Identical protocol to the program’s other instances: bars committed before graded runs, manipulation checks that are bars, a coded cooling-off, disjoint graded seeds, kept failures. **[TODO/D8-V1: state the D8-V1 prereg’s UP-phase responsiveness gate, change-point rule, and the eight screened live beacon targets as sealed.]**

IV. SEALED CONTROL-PLANE CELLS

Table I gives the sealed, graded control-plane results; these are the data that is in.

The ordering $0.351 > 0.184 > 0.083$ tracks protocol opacity (path-vector churn is less visible on the update channel than link-state floods); within the link-state family OSPF and IS-IS differ by $2.2\times$ at matched normalization, placed by implementation evidence-pacing. The BMP cell validates the witness channel itself: on that substrate the retrofit buys visibility structure ($2\times$ events, 137 ms vs. 5.18 s p95), and a measured falsity gap needs churn that never surfaces on the update channel (policy/damping)—the wild’s lane, which the dataplane cell enters.

TABLE I
SEALED CONTROL-PLANE CELLS (GRADED, COMMITTED AS EXECUTED). THE 60 s QUIESCENCE CERTIFICATE’S FALSE-CLEAR IS ORDERED BY PROTOCOL OPACITY.

Cell	protocol	60 s FC	verdict
XPROTO-BGP	BGP (path-vector)	0.351	PASS
XPROTO-ISIS	IS-IS (link-state)	0.184	PASS
XPROTO-OSPF	OSPF (link-state)	0.083	PASS
XPROTO-BMP	BMP	validated	PASS/FAIL kept
	witness-grading		

V. THE DATAPLANE-JOIN CELL (D8-V1)

[TODO/D8-V1: This is the decisive cell and the reason the paper is gated. Fill from the sealed D8-V1 graded record once RIPE Atlas credits reach the launch threshold and the eight beacon measurements run: the live-reachability false-clear of the quiescence certificate vs. the dataplane witness, the consumer-relative spread by vantage, and the confirmation (or refutation) of the 30–120 s posterior-detector dominance band. If the band is refuted, the paper’s product claim reverts to control-plane only and says so.]

VI. THE POSTERIOR-DETECTOR DOMINANCE BAND

The 30–120 s band is where a calibrated posterior detector dominates the naive quiescence monitor. It was V0-discovered (the sealed B3 bar failed and the band is where the two wins landed); its status is *confirmation pending*. **[TODO/D8-V1: promote to a confirmed claim iff D8-V1 passes; otherwise keep the honest “discovered, not confirmed” wording.]**

VII. DISCUSSION AND RELATED WORK

Age of information [5] bounds update staleness; consumer-relative false-clear measures the reachability consequence. The certificate/witness grammar is shared with the database instance (replication-log witness) and the radio-access instance (HARQ witness); routing contributes the *opacity-orders-the-band* finding. **[TODO/D8-V1: position vs. specific convergence-monitoring and BGP-anomaly-detection literature after the D8-V1 result fixes the headline.]**

VIII. CONCLUSION

[TODO/D8-V1: Write after D8-V1. The control-plane cells establish opacity-ordered quiescence vacuity; the dataplane cell is required to claim the live-reachability result and the product surface.]

REPRODUCIBILITY

Pre-registrations, verdict runners, and graded records are archived with the program [6]; the dataplane cell uses the RIPE Atlas network [7].

REFERENCES

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